

* Complex Analysis *

- Euler's formula, De Moivre's theorem, Roots of complex numbers, Functions of complex variables
 - Analyticity & Cauchy-Riemann conditions
 - Singular functions:
 - Poles
 - Branch points
 - Order of singularity
 - Branch cuts
 - Integration of a function of a complex variable
 - Cauchy's inequality
 - Cauchy's integral formula
 - Simply and multiply connected regions
 - Laurent and Taylor series expansions
 - Residues & Residue theorem
 - Application in solving definite integrals
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* Definition $\rightarrow$ A complex number z is an ordered pair of real numbers : $z \equiv (x, y)$ with addition and multiplication defined as follows.

For $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$

$$z_1 + z_2 \equiv (x_1 + x_2, y_1 + y_2)$$

$$z_1 z_2 \equiv (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2)$$

Numbers of the form $(x, 0)$ are "called" real numbers and those of the form $(0, y)$ are called imaginary numbers.

$$(x, 0) + (0, y) = (x, y) = (x, 0) + (0, 1)(y, 0)$$

$$\operatorname{Re} z = x, \operatorname{Im} z = y$$

$$z_1 = z_2 \text{ iff } x_1 = x_2 \text{ and } y_1 = y_2$$

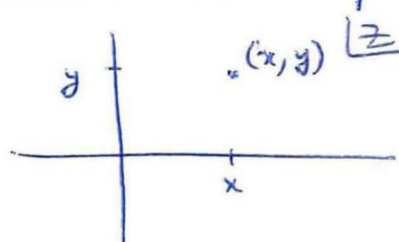
$$(0,1) \equiv i, \quad z = x + iy \quad \text{where } x \equiv (x,0)$$

$$i^2 = (0,1) \times (0,1) = -1$$

- Closure under addition & multiplication
- Additive & multiplicative identity (0 & 1)
- Additive inverse $\forall z$ $(-z)$
- Multiplicative inverse $\forall z$ except 0. $(1/z)$
- Closure under exponentiation (defined later) $z_1^{z_2}$

Ex: Find the real & imaginary parts of $1/z$

Geometric Interpretation



Modulus

$$|z| = \sqrt{x^2 + y^2} \quad (\text{Distance from origin})$$

Ex:

$$|z_1 - z_2| = ? \quad (\text{Distance b/w } z_1 \text{ \& } z_2)$$

Ex:

Equation of a circle?

$$|z - z_0| = R$$

Complex conjugate: $\bar{z} = x - iy$

$$z\bar{z} = |z|^2$$

$$z + \bar{z} = 2\operatorname{Re} z$$

$$z - \bar{z} = 2i\operatorname{Im} z$$

$$\left[\begin{array}{l} \text{Region inside} \\ \text{a circle:} \\ |z - z_0| \leq R \end{array} \right]$$

Further properties of moduli

$$|\bar{z}| = |z|$$

$$|z_1 z_2| = |z_1| |z_2|$$

$$|z|^2 = (\operatorname{Re} z)^2 + (\operatorname{Im} z)^2$$

$$|z| \geq |\operatorname{Re} z| \geq \operatorname{Re} z, \quad |z| \geq |\operatorname{Im} z| \geq \operatorname{Im} z$$

$$|z_1 + z_2| \leq |z_1| + |z_2| \quad (\text{Triangle Inequality})$$

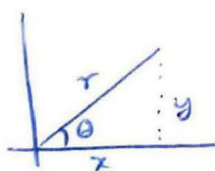
Proof: $|z_1 + z_2|^2 = (z_1 + z_2)(\overline{z_1 + z_2})$

$$\begin{aligned} &= |z_1|^2 + |z_2|^2 + z_1 \bar{z}_2 + \bar{z}_1 z_2 \\ &= |z_1|^2 + |z_2|^2 + 2 \operatorname{Re} z_1 \bar{z}_2 \\ &\leq |z_1|^2 + |z_2|^2 + 2 |z_1| |z_2| \\ &= (|z_1| + |z_2|)^2 \end{aligned}$$

$$\Rightarrow |z_1 + z_2| \leq |z_1| + |z_2| \quad \because |z| \geq 0$$

Polar form ~~(Euler's identity formula)~~

$(r, \theta) \rightarrow$ Polar coordinates corresponding to (x, y)



$$r = |z|$$

$$\theta = \tan^{-1} \frac{y}{x} = \arg z$$

Principal value of $\arg z \equiv \operatorname{Arg} z \in (-\pi, \pi]$

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2$$

Exponential Form / Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta \equiv \text{cis } \theta$$

$$e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}$$

$$e^{i2n\pi} = 1 \quad n = 0, \pm 1, \pm 2, \dots$$

From Taylor series expansion of $\cos$ & $\sin$

$$\begin{aligned} \cos \theta + i \sin \theta &= 1 - \frac{\theta^2}{2!} + \dots + i\left(\theta - \frac{\theta^3}{3!} + \dots\right) \\ &= 1 + i\theta + \frac{(i\theta)^2}{2!} + \dots \\ &= e^{i\theta} \end{aligned}$$

$$\boxed{z_1 = z_2 \Rightarrow r_1 = r_2 \text{ and } \theta_1 = \theta_2 + 2k\pi, k \in \mathbb{Z}}$$

Parametric representation of a circle

$$z = z_0 + R e^{i\theta} \quad 0 \leq \theta < 2\pi$$

$$(e^{i\theta})^n = e^{in\theta}$$

$$\Rightarrow (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad \text{de Moivre's formula}$$

n^{th} roots of unity

$$z^n = 1$$

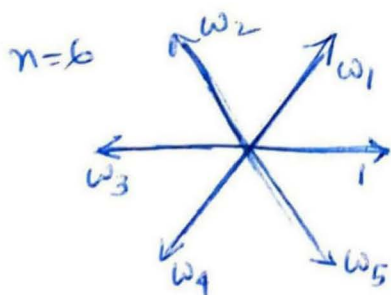
$$(r e^{i\theta})^n = 1 \Rightarrow r^n e^{in\theta} = 1 e^{i0}$$

$$\Rightarrow r^n = 1 \Rightarrow r = 1$$

$$n\theta = 2k\pi \Rightarrow \theta = \frac{2k\pi}{n}$$

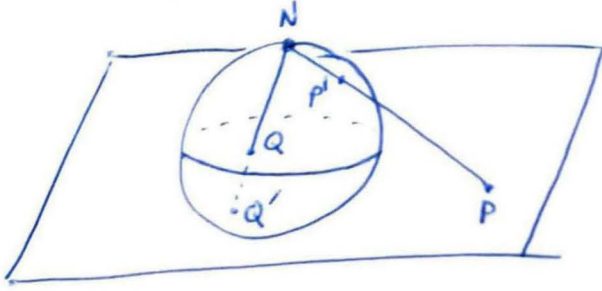
$\Rightarrow n$ distinct roots:

$$z = \exp\left(i \frac{2k\pi}{n}\right) \quad k = 0, 1, 2, \dots, (n-1)$$



Stereographic projection

Complex plane $\rightarrow$ Surface of sphere of unit radius



North pole $\rightarrow$ Point at infinity

South pole $\rightarrow$ origin

Lec 1+2

25/01

Lec Hrs - 2

Ex:

① Show $\frac{1+2i}{3-4i} + \frac{2-i}{5i} = -\frac{2}{5}$

② $(1-i)^4 = -4$

③ $|(2\bar{z}+5)(\sqrt{2}-i)| = \sqrt{3}|(2z+5)|$

④ Use de Moivre's formula to derive

① $\cos 3\theta = \cos^3\theta - 3\cos\theta \sin^2\theta$

② $\sin 3\theta = 3\cos^2\theta \sin\theta - \sin^3\theta$

⑤ Establish $1+z+z^2+\dots+z^n = \frac{1-z^{n+1}}{1-z}$

From this derive Lagrange's trigonometric identity

$$1+\cos\theta+\cos 2\theta+\dots+\cos n\theta = \frac{1}{2} + \frac{\sin[(n+\frac{1}{2})\theta]}{2\sin(\theta/2)}$$

⑥ Sketch the regions

① $|z-2+i| \leq 1$

② $|z-4| \geq |z|$

Function of a complex variable

$$f(z) = u(x, y) + i v(x, y) = w$$

Graphical Representation

- ① Plot surfaces $u(x, y)$ & $v(x, y)$ on x, y plane
- ② Plot u, v on w plane

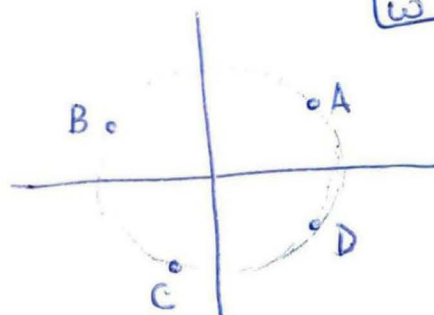
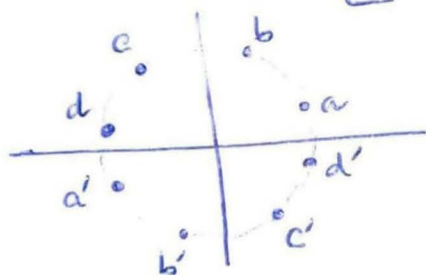
Function — mapping from z -plane to w -plane

E.g. $f(z) = z^2 = (x+iy)^2$
 $= x^2 - y^2 + 2ixy$

$$u(x, y) = x^2 - y^2$$

$$v(x, y) = 2xy$$

z



LIMIT: " $z \rightarrow z_0$ "

Infinite sequence of point approaching z_0 along any path in z -plane

Continuity at z_0 :

$f(z_0)$ exists and $\lim_{z \rightarrow z_0} f(z) = f(z_0)$

w

Upper half z -plane maps onto the entire w -plane
 $\hookrightarrow$ same for lower half z -plane

Inverse mapping of $z^2 \rightarrow z^{1/2}$ looks multivalued.

Convert to single valued by enlarging the domain.

The enlarged domain is called the Riemann surface.

For the square root function, the Riemann surface consists of 2 copies of the complex plane (Riemann sheets)

On sheet 1, z is taken as $z = r e^{i\theta}$ with $0 \leq \theta < 2\pi$

On sheet 2, $z = r e^{i\theta}$ with $2\pi \leq \theta < 4\pi$.

The sheets are connected along an infinitely long line going from $z=0$ to the point at infinity in any direction. On crossing this line, one goes from the first sheet to the second and vice versa.

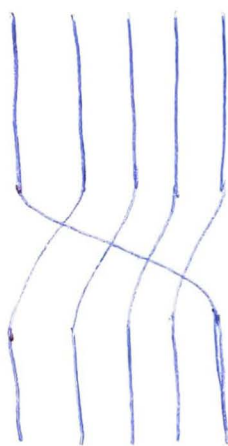
The points $z=0$ & $z=\infty$ are called branch points and the above line is called a branch cut.

The branch cut can be curved as well. It should be non-intersecting with itself and go from $z=0$ to $z=\infty$.

Thus, the square root function is continuous ~~on the~~ for $0 < |z| < \infty$ when the domain is a two-sheeted Riemann surface.

For $f(z) = z^{1/5}$, we need a 5-sheeted Riemann surface

Edge view:
across the
branch cut



Differentiability:

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

Must exist
and be unique

~~A function is ANALYTIC at z if it has a derivative there.~~

A function is analytic in an open region R if it is differentiable at all points $z \in R$. Also called "regular".

①

Quantization of Energy levels

Consider two-state Hamiltonian

$$H_0 = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

Time independent Schrödinger equation

$$H_0 |\psi\rangle = E |\psi\rangle$$

Eigenstates: $|\psi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|\psi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Eigenvalues: $E_1 = a$, $E_2 = b$

Interaction with external field: $H_1 = z \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

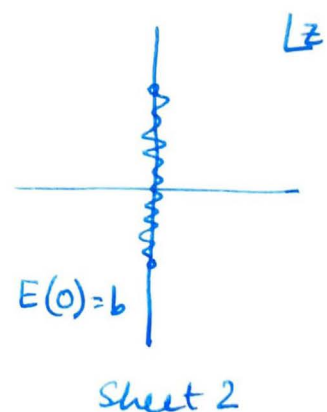
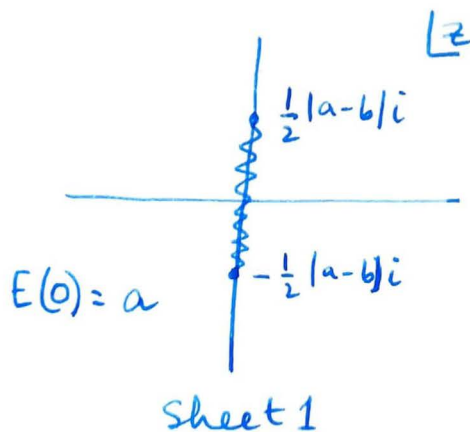
↓
models electric charge etc
and field strength

$$H = \begin{pmatrix} a & z \\ z & b \end{pmatrix}$$

$$H |\psi\rangle = E |\psi\rangle$$

$$\begin{vmatrix} a-E & z \\ z & b-E \end{vmatrix} = 0 \Rightarrow E(z) = \frac{a+b}{2} \pm \sqrt{\left(\frac{a-b}{2}\right)^2 + z^2}$$

$E(z)$ defined on 2-sheeted Riemann surface with
branch points at $\pm \frac{1}{2}(a-b)i$



Path 1 $\Delta z = \Delta x$ (real)

$$f'(z) = \lim_{\Delta x \rightarrow 0} \frac{u(x+\Delta x, y) + iv(x+\Delta x, y) - u(x, y) - iv(x, y)}{\Delta x}$$
$$= u_x(x, y) + i v_x(x, y)$$

Path 2 $\Delta z = i \Delta y$ (purely imaginary)

$$f'(z) = \lim_{\Delta y \rightarrow 0} \frac{u(x, y+\Delta y) + iv(x, y+\Delta y) - u(x, y) - iv(x, y)}{i \Delta y}$$
$$= -i u_y(x, y) + v_y(x, y)$$

Uniqueness $\Rightarrow$ $u_x = v_y$ & $u_y = -v_x$ Cauchy-Riemann Conditions

Taking another derivative on both sides of above equations, we get,

$$\left. \begin{array}{l} u_{xx} + u_{yy} = 0 \\ \text{and } v_{xx} + v_{yy} = 0 \end{array} \right\} \begin{array}{l} u \text{ \& } v \text{ are "harmonic" functions} \\ \text{(satisfy Laplace's equation)} \end{array}$$

← Lec. Hrs 3 & 4

(A) Points ~~at~~ where $f(z)$ is not differentiable are called singular points or singularities of $f(z)$

Real & Imaginary parts of an analytic function ~~are~~ form a conjugate pair of harmonic functions.

Ex: Given $u(x, y) = x^2 - y^2$, find $f(z)$ if it is analytic

Check harmonicity: $u_{xx} = 2$, $u_{yy} = -2$ ✓

$$v_y = u_x = 2x \Rightarrow v(x, y) = 2xy + g(x)$$

$$v_x = -u_y = 2y = 2y + g'(x) \Rightarrow g'(x) = 0 \Rightarrow g(x) = C$$

$$\therefore f(z) = x^2 - y^2 + 2ixy + C = z^2 + C$$

$$f(z) = R(z) e^{i\theta(z)}$$

$$z = x + iy = r e^{i\theta}$$

$$\begin{aligned} \text{I) } f'(z) &= \lim_{\Delta x \rightarrow 0} \frac{R(x+\Delta x, y) e^{i\theta(x+\Delta x, y)} - R(x, y) e^{i\theta(x, y)}}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{[R(x, y) + \Delta x R_x(x, y) + O(\Delta x^2)] e^{i[\theta(x, y) + \Delta x \theta_x(x, y)]} - R e^{i\theta}}{\Delta x} \end{aligned}$$

[Notation: No arguments $\Rightarrow$ arguments (x, y)]

$$= \lim_{\Delta x \rightarrow 0} \frac{(R + \Delta x R_x) e^{i\theta} (1 + \Delta x i \theta_x) - R e^{i\theta}}{\Delta x}$$

$$= R_x e^{i\theta} + i R \theta_x e^{i\theta}$$

$$\text{II) } f'(z) = \lim_{\Delta y \rightarrow 0} \frac{R(x, y+\Delta y) e^{i\theta(x, y+\Delta y)} - R e^{i\theta}}{i \Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{(R + \Delta y R_y) e^{i\theta} (1 + i \Delta y \theta_y) - R e^{i\theta}}{i \Delta y}$$

$$= R \theta_y e^{i\theta} - i R_y e^{i\theta}$$

From I & II,

$$R_x = R \theta_y \quad \text{and} \quad R_y = -R \theta_x$$

$$\text{III) } f'(z) = \lim_{\Delta r \rightarrow 0} \frac{R(r+\Delta r, \theta) e^{i\theta(r+\Delta r, \theta)} - R e^{i\theta}}{\Delta r e^{i\theta}}$$

$$= \lim_{\Delta r \rightarrow 0} \frac{(R + \Delta r R_r) e^{i\theta} (1 + i \Delta r \theta_r) - R e^{i\theta}}{\Delta r e^{i\theta}}$$

$$= R_r e^{i(\theta-\theta)} + i R \theta_r e^{i(\theta-\theta)}$$

$$\text{IV) } f'(z) = \lim_{\Delta \theta \rightarrow 0} \frac{R(r, \theta+\Delta \theta) e^{i\theta(r, \theta+\Delta \theta)} - R e^{i\theta}}{r i e^{i\theta} \Delta \theta}$$

$$= \lim_{\Delta \theta \rightarrow 0} \frac{(R + \Delta \theta R_\theta) e^{i\theta} (1 + i \Delta \theta \theta_\theta) - R e^{i\theta}}{r i e^{i\theta} \Delta \theta}$$

$$= \frac{R \theta_\theta}{r} e^{i(\theta-\theta)} - i \frac{R_\theta}{r} e^{i(\theta-\theta)}$$

From (III) & (IV),

$$R_r = \frac{R \Theta_\theta}{r} \quad \text{and} \quad R_\theta = -r R \Theta_r$$

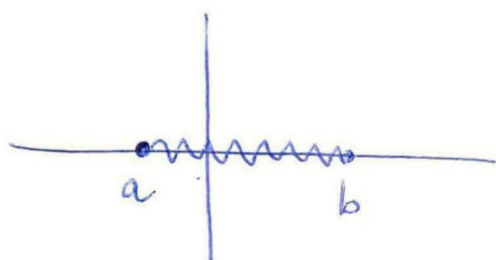
Log $f(z) = \log z$ $z = re^{i\theta}$
 $= \log r + i\theta$

Riemann surface $\rightarrow$ Infinite number of sheets each of which map to one horizontal strip of width 2π in the Li-plane

~~0~~ For the k^{th} sheet, $2\pi k < \theta \leq 2\pi(k+1)$
 $k \in \mathbb{Z}$

$\sqrt{(z-a)(z-b)}$

For $|z| \rightarrow \infty$, $f(z) \sim z$ or $f(z) \sim -z$
 No branch point at ∞ .



2-sheeted Riemann surface

Example of disconnected sheets

$$f(z) = i^z = e^{z \log i} = e^{z \log e^{i\pi/2 + 2n\pi i}} \\ = e^{iz(2n+1/2)\pi}$$

There is no branch point, no branch cut

Ex: (From NET book, pg 132, 10)

- Show that $u(x,y) = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$ is harmonic.
 Determine the analytic function $f(z)$ whose real part $u(x,y)$ is.

Analytic/Regular/Holomorphic $\Rightarrow$ Differentiable and single-valued
(at least on a Riemann sheet) at all points in a region R .

Complex Path Integrals / Contour Integrals

$$\int_C dz f(z)$$

Example 1: Find $\int_C |z| dz$ where C is the path that runs from $z = -1$ to $z = +1$ along (a) Real axis (b) Unit circle clockwise.

a) $\int_C |z| dz = \int_{-1}^1 |t| dt \quad z = t, t \in \mathbb{R}$

$$= 2 \int_0^1 t dt = 1$$

b) $z = e^{i\theta} \quad \theta: \pi \rightarrow 0 \quad dz = ie^{i\theta} d\theta$

$$\Rightarrow \int_C |z| dz = \int_{\pi}^0 ie^{i\theta} d\theta = e^{i\theta} \Big|_{\pi}^0 = 1 + 1 = 2$$

Example 2: Find $\int_C z^2 dz$ along same paths as above. (Ans. = $2/3$)

Cauchy's Theorem: If $f(z)$ is analytic in a ^{simply connected} region that bounds a closed curve C , then, $\oint_C f(z) dz = 0$

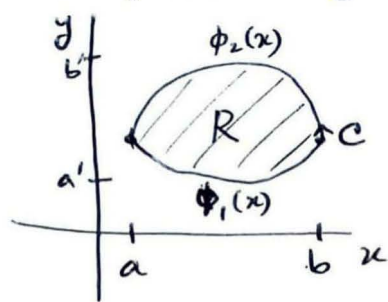
Consequence: $\int_{z_1}^{z_2} f(z) dz$ is path independent for all paths that connect z_1 and z_2 and $f(z)$ is analytic in a ^{simply connected} region that bounds these paths.

Entire function: Analytic $\forall z$ with $|z| < \infty$

If $f(z)$ is analytic $\forall z$ (including $z = \infty$), $f(z) = \text{constant}$.

Simply connected region — ① Includes boundaries ② Any two points can be connected by a continuous path ③ Any closed curve can be smoothly shrunk to a point.

Proof of Cauchy's Theorem



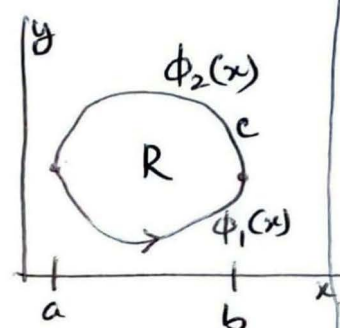
$$I = \oint_C f(z) dz$$

$$= \oint_C (dx + i dy) (u + i v)$$

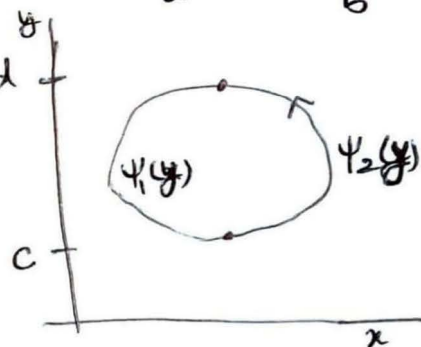
Green's Theorem:

$$\oint [dx P(x, y) + dy Q(x, y)] = \int_R dx dy (Q_x - P_y) \dots \textcircled{1}$$

$$\int_a^b dx \int_{\phi_1(x)}^{\phi_2(x)} dy P_y = \int_a^b dx [P(x, \phi_2(x)) - P(x, \phi_1(x))] = - \oint dx P(x, y)$$



$$\int_c^d dy \int_{\psi_1(y)}^{\psi_2(y)} dx Q_y = \int_c^d dy [Q(\psi_2(y), y) - Q(\psi_1(y), y)] = \oint dy Q(x, y)$$



$\Rightarrow \textcircled{1}$

$$\Rightarrow I = \oint \cancel{dx} [(u + i v) dx + (i u - v) dy]$$

$$= \int_R dx dy [i u_x - v_x - u_y - i v_y]$$

$= 0$ by Cauchy Riemann conditions.

Morera's Theorem : Path independence of integrals in a simply connected region $R \Rightarrow$ and analyticity in R .

Cauchy's Integral Theorem

$$f(z_0) = \frac{1}{2\pi i} \oint_C dz \frac{f(z)}{z-z_0}$$

$f(z)$ analytic
in simply connected
region around C

$$\left[\oint_C \frac{dz}{z} = \log z \right]_{\substack{z=1 \text{ on sheet } n+1 \\ z=1 \text{ on sheet } n}} = 2\pi(n+1)i - 2\pi n i = 2\pi i$$

$\oint_C \frac{dz}{z} = 2\pi i$

$\Rightarrow \oint \frac{dz}{z-z_0} = 2\pi i$

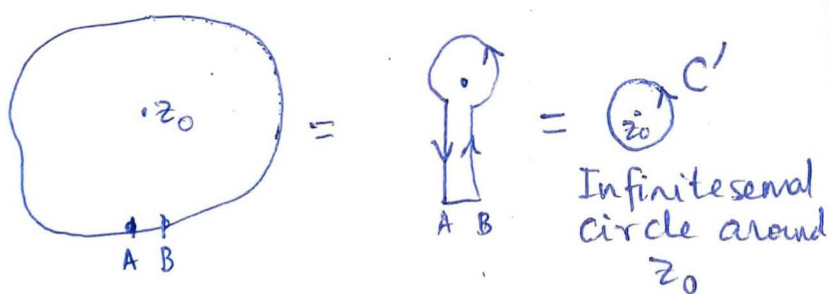


Proof:

$$I = \underbrace{\oint_C \frac{1}{2\pi i} dz \frac{f(z_0)}{z-z_0}}_{I_1} + \underbrace{\frac{1}{2\pi i} \oint_C dz \frac{f(z)-f(z_0)}{z-z_0}}_{I_2}$$

$$I_1 = f(z_0) \times \frac{1}{2\pi i} \oint_C \frac{dz}{z-z_0} = f(z_0)$$

For I_2 :



On C' , $\frac{f(z)-f(z_0)}{z-z_0} = f'(z_0)$

$$\Rightarrow I_2 = \frac{f'(z_0)}{2\pi i} \oint_{C'} dz$$

$= 0 \quad \because \text{Length of } C' \rightarrow 0$

$$\Rightarrow I_2 = 0 \quad \& \quad I = f(z_0) \quad \checkmark$$

$$\left| \int_C dz f(z) \right| \leq \int_C |dz| |f(z)| \leq (\text{Length of path } C) \times \max_C |f(z)|$$

Problems from NET Book

① Find the residue of the following functions at ∞ infinity

a) $f(z) = \frac{z}{(z-a)(z-b)}$

b) $f(z) = \frac{z^3 - z^2 + 1}{z^3}$

② Given $\int_0^\infty e^{-(a+ib)x} dx = \frac{1}{a+ib}$, $a > 0$, Show $\int_0^\infty \frac{\sin bx}{x} dx = \frac{\pi}{2}$

③ (a) Find rad. of convergence of the series $\sum_{n=1}^\infty \frac{n!}{n^n} x^n$

(b) Prove by method of residues $\int_{-\infty}^\infty \frac{\cos mx}{(x+c)^2 + a^2} dx = \frac{\pi}{a} e^{-a|m|} \cos cm$
 $a, c \in \mathbb{R}$ and $a > 0$

~~$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z) dz}{z - z_0}$~~

$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z) dz}{z - z_0}$

$f(z)$ analytic in simply connected region around C .

$\Rightarrow f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z) dz}{(z - z_0)^2}$, $f''(z_0) = \frac{2!}{2\pi i} \oint_C \frac{f(z) dz}{(z - z_0)^3}$,

.... $\boxed{f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z) dz}{(z - z_0)^{n+1}}}$ $\Rightarrow$ Derivatives of all orders exist at z_0 .

$\Rightarrow$ Existence of Taylor series expansion at z_0 .

$f(z) = \sum_{n=0}^\infty a_n (z - z_0)^n$ where $a_n = f^{(n)}(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z) dz}{(z - z_0)^{n+1}}$

Consider $f(z)$ which is analytic in the region $|z| < R$.

$\Rightarrow f(z) = \sum_{n=0}^\infty a_n z^n$ where $a_n = \frac{1}{2\pi i} \oint_C \frac{f(z) dz}{z^{n+1}}$

$\therefore |a_n| < \frac{1}{2\pi} \frac{2\pi r}{r^{n+1}} \max_{|z|=r} |f(z)|$

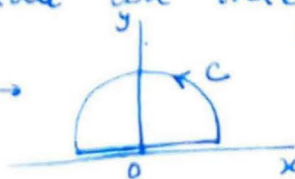
Choose C as a circular path of radius $r < R$ around 0 .

$$\Rightarrow |a_n| < r^{-n} \max_{|z|=r} |f(z)| \quad (\text{Cauchy's Inequality})$$

For a bounded entire function, r can be chosen to be arbitrarily large $\Rightarrow a_n = 0$ for $n \geq 1$ for a bounded entire function $\Rightarrow$ Liouville's theorem: A bounded entire function must be a constant.

Using Cauchy's theorem to calculate an integral:

$$I = \oint_C \frac{dz}{z^2 + 1} \quad \text{where } C \text{ is } \rightarrow \text{ (Flat part on the real axis)}$$



$$= \oint_C \frac{dz}{(z+i)(z-i)}$$

$$= \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-i} dz \quad \text{where } f(z) = \frac{2\pi i}{z+i} \text{ which is analytic in a simply connected region containing } C$$

$$= f(i)$$

$$= \pi$$

Derivation of Taylor Series:

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)}{w-z} dw$$

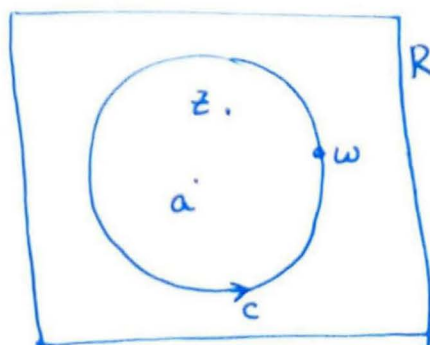
$f(z)$ is analytic in R

Choose C to be a circular contour

centered at ' a ' and of radius $r = |w-a|$ such that $|w-a| > |z-a|$

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(w) dw}{(w-a)(1-E)} \quad \text{where } E = 1 - \frac{w-z}{w-a} = \frac{z-a}{w-a}$$

$$\frac{1}{1-E} = 1 + E + E^2 + \dots + E^N + \frac{E^{N+1}}{1-E}$$



Term-by-term operations

Within its circle of convergence it is okay to differentiate and integrate a Taylor series term-by-term.

If we have another Taylor series which converges in that circle, it is ~~okay~~ okay to add the two series or multiply them together.

$$f_1(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n, \quad f_2(z) = \sum_{n=0}^{\infty} b_n (z-z_0)^n$$

$$f_1(z) + f_2(z) = \sum_{n=0}^{\infty} (a_n + b_n) (z-z_0)^n$$

$$f_1(z) f_2(z) = \sum_{n_1=0}^{\infty} a_{n_1} (z-z_0)^{n_1} \sum_{n_2=0}^{\infty} b_{n_2} (z-z_0)^{n_2}$$

$$= \sum_{n=0}^{\infty} c_n (z-z_0)^n, \quad c_n = \sum_{m=0}^n a_m b_{n-m}$$

$$f_1'(z) = \sum_{n=1}^{\infty} n a_n (z-z_0)^{n-1}$$

$$\int f_1(z) dz = \text{constant} + \sum_{n=0}^{\infty} \frac{a_n}{n+1} (z-z_0)^{n+1}$$

If $a_0 \neq 0$,

$$\begin{aligned} \frac{1}{f_1(z)} &= \frac{1}{a_0 + \sum_{n=1}^{\infty} a_n (z-z_0)^n} = \frac{1}{a_0} \left[1 + \frac{1}{a_0} \sum_{n=1}^{\infty} a_n (z-z_0)^n \right]^{-1} \\ &= \frac{1}{a_0} - \frac{a_1}{a_0^2} (z-z_0) + \frac{a_1^2 - a_0 a_2}{a_0^3} (z-z_0)^2 + \frac{2a_0 a_1 a_2 - a_1^3 - a_0^2 a_3}{a_0^4} (z-z_0)^3 + \dots \end{aligned}$$

Inverting a function, in general, results in a change in the radius of convergence (introduces new singularities).

Uniqueness of Taylor series

Suppose $f(z)$ is represented by two Taylor series around the point $z=z_0$

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n \quad \text{and} \quad f(z) = \sum_{n=0}^{\infty} b_n (z-z_0)^n$$

Differentiating the first series m times and setting $z=z_0$, we get,

$$f^{(m)}(z_0) = m! a_m$$

Similarly, from the second series, we get,

$$f^{(m)}(z_0) = m! b_m$$

Thus, $a_m = b_m \quad \forall m \Rightarrow$ Taylor series is unique.

$\Rightarrow$ If we can find a ^{convergent} series that is of the form $\sum_{n=0}^{\infty} a_n (z-z_0)^n$ representing the function around $z=z_0$, then ~~it~~ it must be its Taylor series about that point.

$\hookrightarrow$ The same is true for a Laurent series in its region of convergence.

Regions of convergence of a Taylor series from tests of convergence of series

$$S = \sum_{n=0}^{\infty} A_n, \quad f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n$$

① Ratio test: Series S converges absolutely if

$$\lim_{n \rightarrow \infty} \left| \frac{A_{n+1}}{A_n} \right| < 1$$

and diverges if $\lim_{n \rightarrow \infty} \left| \frac{A_{n+1}}{A_n} \right| > 1$

$\Rightarrow$ Taylor series converges if

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1} (z-z_0)^{n+1}}{a_n (z-z_0)^n} \right| < 1$$

$$\Rightarrow |z-z_0| < r, \text{ where } r = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|$$

② Root Test: $\lim_{n \rightarrow \infty} |A_n|^{1/n} \begin{cases} < 1 \Rightarrow \text{convergence} \\ > 1 \Rightarrow \text{divergence} \end{cases}$

~~For~~ Taylor series converges if

$$\lim_{n \rightarrow \infty} \left| a_n (z-z_0)^n \right|^{1/n} < 1$$

$$\Rightarrow |z-z_0| < r = \lim_{n \rightarrow \infty} |a_n|^{-1/n}$$

$$\Rightarrow f(z) = \frac{1}{2\pi i} \oint \frac{f(w)}{w-a} \left[1 + \epsilon + \epsilon^2 + \dots + \epsilon^N + \frac{\epsilon^{N+1}}{1-\epsilon} \right] dw$$

$$= \sum_{n=0}^N (z-a)^n \frac{1}{2\pi i} \oint \frac{f(w) dw}{(w-a)^{n+1}} + R_N$$

$$= \sum_{n=0}^N (z-a)^n \frac{f^{(n)}(a)}{n!} + R_N$$

$$R_N = \frac{1}{2\pi i} \oint \frac{f(w) \epsilon^{N+1}}{w-z} dw$$

$$\Rightarrow |R_N| \leq \frac{1}{2\pi} \left(\max_c \frac{|f(w)|}{|w-z|} \right) 2\pi r |\epsilon|^{N+1}$$

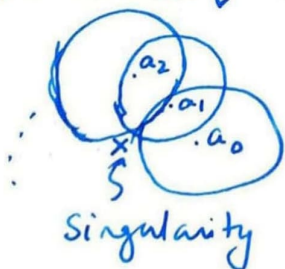
$$\leq \frac{1}{2\pi} \frac{\max_c |f(w)|}{\min_c |w-z|} r |\epsilon|^{N+1}$$

$$= \frac{\max_c |f(w)|}{r - |z-a|} r |\epsilon|^{N+1} \rightarrow 0 \text{ as } N \rightarrow \infty$$

$$\Rightarrow R_N \rightarrow 0 \text{ as } N \rightarrow \infty \Rightarrow f(z) = \sum_{n=0}^{\infty} (z-a)^n \frac{f^{(n)}(a)}{n!}$$

The above derivation works as long as the point z is closer to the point a , than the nearest singularity of $f(z)$ to point ' a '. Thus, the radius of convergence of a Taylor series is the distance from the nearest singularity.

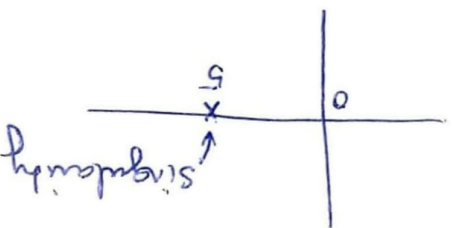
[Analytic continuation (not in syllabus): Going around a singularity using ~~the~~ Taylor series a sequence of Taylor series centered at ~~var~~ varying points.



Laurent Series: $f(z) = \sum_{n=-\infty}^{\infty} a_n (z-z_0)^n \rightarrow$ Needs $f(z)$ to be analytic in an annular region between 2 concentric circles centered at z_0 .

Laurent Series Examples

① Expand $f(z) = \frac{z}{(z-5)^5}$ about $z=5$ in a Laurent series.



Region of convergence: $0 < |z-5| < \infty$

$$= \frac{z-5+5}{z-5+5} \cdot \frac{(z-5)^5}{(z-5)^5} + \frac{1}{(z-5)^4}$$

$$f(z) = \frac{z}{(z-5)^5}$$

② about $z=5$

$$\frac{z}{z-5+5} = \frac{z+2}{z-5+5}$$

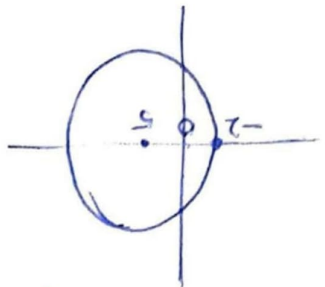
$$= \frac{z-5+5}{z-5+5} \left[1 + \frac{z-5}{5} \right]$$

$$= \left[1 - \frac{z-5}{5} + \frac{(z-5)^2}{49} - \frac{(z-5)^3}{343} + \dots \right] \left[1 + \frac{z-5}{5} \right]$$

$$= 1 + \frac{z-5}{5} + \frac{z-5}{49-35} + \frac{(z-5)^2}{49 \times 5 - 343} + \dots$$

$$= 1 - \frac{z-5}{2} + \frac{(z-5)^2}{14} - \frac{(z-5)^3}{98} + \dots$$

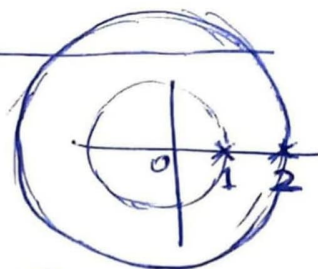
Region of convergence: $7 < |z-5| < \infty$



Note: The same function is analytic in the circle $|z-5| < 7$ and must have a Taylor series in that region.

$$\begin{aligned}
 \frac{z}{z+2} &= \frac{z-5+5}{7+z-5} \\
 &= \frac{z-5+5}{7} \left[1 + \frac{z-5}{7} \right]^{-1} \\
 &= \left(\frac{5}{7} + \frac{z-5}{7} \right) \left(1 - \frac{z-5}{7} + \frac{(z-5)^2}{49} - \frac{(z-5)^3}{343} + \dots \right) \\
 &= \frac{5}{7} + \left(\frac{1}{7} - \frac{5}{49} \right) (z-5) + \left(-\frac{1}{49} + \frac{5}{343} \right) (z-5)^2 + \dots \\
 &= \frac{5}{7} + \frac{2}{49} (z-5) - \frac{2}{343} (z-5)^2 + \dots
 \end{aligned}$$

③ $f(z) = \frac{1}{(z-1)(2-z)}$ around $z=0$



(a) $|z| < 1 \Rightarrow f(z)$ is analytic $\Rightarrow$ Taylor series

~~$f(z) = \frac{1}{2} \left(\frac{1}{1-z} - \frac{1}{1-\frac{z}{2}} \right)$~~

$$\begin{aligned}
 f(z) &= -\frac{1}{2} (1-z)^{-1} \left(1 - \frac{z}{2} \right)^{-1} \\
 &= -\frac{1}{2} (1+z+z^2+\dots) \left(1 + \left(\frac{z}{2}\right) + \left(\frac{z}{2}\right)^2 + \dots \right) \\
 &= -\frac{1}{2} - \frac{1}{2} \left(1 + \frac{1}{2} \right) z - \frac{1}{2} \left(1 + \frac{1}{2} + \frac{1}{4} \right) z^2 - \dots \\
 &= -\frac{1}{2} - \frac{3}{4} z - \frac{7}{8} z^2 - \dots
 \end{aligned}$$

$$(b) \quad 1 < |z| < 2 \quad \Rightarrow \quad \left| \frac{1}{z} \right| < 1 \quad \text{and} \quad \left| \frac{z}{2} \right| < 1$$

$$f(z) = \frac{1}{2z} \left(1 - \frac{1}{z}\right)^{-1} \left(1 - \frac{z}{2}\right)^{-1}$$

$$= \frac{1}{2z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right) \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots\right)$$

~~$$= \frac{1}{2z} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right) \left(1 + \frac{z}{2} + \frac{z^2}{4} + \frac{z^3}{8} + \dots\right)$$~~

$$= \frac{1}{2z} \left[\dots + \frac{1}{z^2} \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots\right) \right.$$

$$+ \frac{1}{z} \left(1 + \frac{1}{2} + \frac{1}{4} + \dots\right)$$

$$+ \left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots\right)$$

$$+ z \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots\right)$$

$$+ z^2 \left(\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots\right) + \dots \left.] \right.$$

$$= \frac{1}{2z} \left[\dots + \frac{2}{z^2} + \frac{2}{z} + 2 + z + \frac{z^2}{2} + \dots \right]$$

$$= \dots + \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2} + \frac{z}{4} + \dots$$

$$(c) \quad |z| > 2 \quad \Rightarrow \quad \left| \frac{1}{z} \right| < 1 \quad \& \quad \left| \frac{2}{z} \right| < 1$$

$$f(z) = -\frac{1}{z^2} \left(1 - \frac{1}{z}\right)^{-1} \left(1 - \frac{2}{z}\right)^{-1}$$

$$= -\frac{1}{z^2} \left(1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots\right) \left(1 + \frac{2}{z} + \frac{4}{z^2} + \frac{8}{z^3} + \dots\right)$$

$$= -\left[\frac{1}{z^2} + \frac{3}{z^3} + \frac{7}{z^4} + \dots \right] = -\frac{1}{z^2} - \frac{3}{z^3} - \frac{7}{z^4} - \dots$$

Method 2 (Using Partial Fractions)

$$f(z) = \frac{1}{(z-1)(2-z)} = \frac{1}{z-1} + \frac{1}{2-z}$$

$$\text{For } |z| < 1, \quad \frac{1}{z-1} = -1(1-z)^{-1} = -1 - z - z^2 - z^3 - z^4 - \dots \quad (i)$$

$$\text{For } |z| > 1, \quad \frac{1}{z-1} = \frac{1}{z} \left(1 - \frac{1}{z}\right)^{-1} = \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \dots \quad (ii)$$

$$\text{For } |z| < 2, \quad \frac{1}{2-z} = \frac{1}{2} \left(1 - \frac{z}{2}\right)^{-1} = \frac{1}{2} + \frac{z}{4} + \frac{z^2}{8} + \frac{z^3}{16} + \dots \quad (iii)$$

$$\text{For } |z| > 2, \quad \frac{1}{2-z} = -\frac{1}{z} \left(1 - \frac{2}{z}\right)^{-1} = -\frac{1}{z} - \frac{2}{z^2} - \frac{4}{z^3} - \frac{8}{z^4} - \dots \quad (iv)$$

∴ For $|z| < 1$, using (i) and (iii), we get,

$$f(z) = -\frac{1}{2} - \frac{3z}{4} - \frac{7z^2}{8} - \frac{15z^3}{16} - \dots$$

For $1 < |z| < 2$, using (ii) and (iii), we get,

$$f(z) = \dots + \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2} + \frac{z}{4} + \frac{z^2}{8} + \frac{z^3}{16} + \dots$$

For $|z| > 2$, using (ii) & (iv), we get,

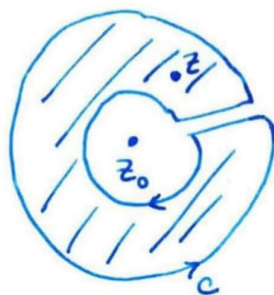
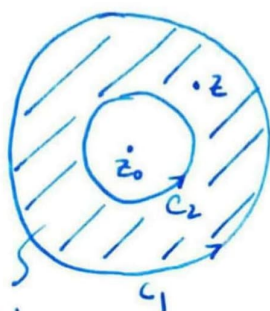
$$f(z) = -\frac{1}{z^2} - \frac{3}{z^3} - \frac{7}{z^4} - \dots$$

MORAL : It is easier to add two similar series than to multiply them. Breaking a complicated function into a sum of simpler functions may be "better" than breaking it into a product of simpler functions.

$$\begin{aligned}\frac{f(z)}{(z-z_0)^{n+1}} &= \sum_{m=-\infty}^{\infty} a_m (z-z_0)^{m-n-1} \\ &= \sum_{p=-\infty}^{\infty} a_{n+p+1} (z-z_0)^p\end{aligned}$$

$$\Rightarrow a_n = \frac{1}{2\pi i} \oint_{C'} \frac{f(z)}{(z-z_0)^{n+1}} dz, \quad C' \text{ is any closed curve encircling } z_0 \text{ and lying in the region where } f(z) \text{ is analytic.}$$

Derivation :



$f(z)$ is analytic here. $\frac{1}{2\pi i} \oint_C \frac{f(w) dw}{(w-z)} = f(z)$

$$\Rightarrow f(z) = \frac{1}{2\pi i} \oint_{C_1} \frac{f(w) dw}{w-z} + \frac{1}{2\pi i} \oint_{C_2} \frac{f(w) dw}{z-w}$$

$$= \frac{1}{2\pi i} \oint_{C_1} \frac{f(w) dw}{(w-z_0)(1-\epsilon_1)} + \frac{1}{2\pi i} \oint_{C_2} \frac{f(w) dw}{(z-z_0)(1-\epsilon_2)}$$

$$\text{where } \epsilon_1 = \frac{z-z_0}{w-z_0}$$

$$\epsilon_2 = \frac{w-z_0}{z-z_0}$$

$$= \frac{1}{2\pi i} \oint_{C_1} \frac{f(w)}{(w-z_0)} \sum_{n=0}^{N_1} \epsilon_1^n dw + \frac{1}{2\pi i} \oint_{C_2} \frac{f(w)}{(z-z_0)} \sum_{n=0}^{N_2-1} \epsilon_2^n dw$$

$$= \sum_{n=-N_2}^{N_1} \frac{(z-z_0)^n}{2\pi i} \oint_D \frac{f(w) dw}{(w-z_0)^{n+1}} + R_1^{(N_1)} + R_2^{(N_2)}$$

where D is any contour around z_0 in the region where $f(z)$ is analytic

$$R_1^{(N_1)} = \frac{1}{2\pi i} \oint_{C_1} \frac{f(\omega)}{\omega - z_0} \frac{\epsilon_1^{N_1+1}}{1 - \epsilon_1} d\omega$$

$$= \frac{1}{2\pi i} \oint \frac{f(\omega)}{\omega - z} \epsilon_1^{N_1+1} d\omega$$

$$\Rightarrow |R_1^{(N_1)}| \leq \frac{1}{2\pi} \oint \frac{|f(\omega)|}{|\omega - z|} |\epsilon_1|^{N_1+1} |d\omega| \rightarrow 0 \Rightarrow R_1^{(N_1)} \rightarrow 0 \text{ as } N_1 \rightarrow \infty$$

$$R_2^{(N_2)} = \frac{1}{2\pi i} \oint_{C_2} \frac{f(\omega)}{z - z_0} \frac{\epsilon_2^{N_2}}{1 - \epsilon_2} d\omega$$

$$\Rightarrow |R_2^{(N_2)}| = \frac{1}{2\pi} \oint \frac{|f(\omega)|}{|z - \omega|} |\epsilon_2|^{N_2} |d\omega| \rightarrow 0 \text{ as } N_2 \rightarrow \infty$$

$$\Rightarrow R_2^{(N_2)} \rightarrow 0 \text{ as } N_2 \rightarrow \infty$$

$\Rightarrow$ Laurent series converges in an annulus around z_0 whose inner and outer radii are optimized so that the function is analytic in the annular region and has singularities just inside the inner circle & just outside the outer circle.

If $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$ in some region around z_0 ,

$$\oint_C f(z) dz = 2\pi i a_{-1}$$

a_{-1} is called the residue of $f(z)$ at z_0 .

Classification of singularities based on Laurent expansion

If inner radius of annular region where the Laurent series converges has zero radius, the singularity is said to be an isolated singularity.

$$\text{If } f(z) \neq \sum_{n=-N}^{\infty} f(z) dz$$

If $f(z) = \sum_{n=-N}^{\infty} a_n (z-z_0)^n$, $a_{-N} \neq 0$

- a) $N \neq 0 \rightarrow$ Removable singularity. (Needs redefinition of $f(z_0)$)
- b) $N=1 \rightarrow$ Simple pole, Residue, $a_{-1} = \lim_{z \rightarrow z_0} f(z)(z-z_0)$
- c) $N \geq 2 \rightarrow$ Pole of order N .
- d) $N=\infty \rightarrow$ Essential singularity

Classification of point at infinity $\Leftrightarrow$ Classification of the point $z=0$ for the function $g(z) = f(1/z)$

Unique ~~over~~ property of $z=\infty$: Can have nonzero residue even when analytic.

$$\oint_{\substack{C \\ \text{around} \\ \infty}} f(z) dz = \oint_{\substack{C \\ \text{around} \\ \text{zero}}} f(1/z') (-1/z'^2) dz'$$

$$\Rightarrow \boxed{\text{Residue of } f(z) \text{ at } \infty = \text{Residue of } \frac{-f(1/z)}{z^2} \text{ at } z=0}$$

Residue at poles

$$f(z) = \sum_{n=-N}^{\infty} a_n (z-z_0)^n$$

$$\Rightarrow f(z)(z-z_0)^N = \sum_{n=-N}^{\infty} a_n (z-z_0)^{n+N} = \sum_{p=0}^{\infty} a_{p-N} (z-z_0)^p$$

a_{-1} is the coefficient of ~~$N-1$~~ $(z-z_0)^{N-1}$ in above Taylor series

$$\Rightarrow \boxed{a_{-1} = \frac{1}{(N-1)!} \lim_{z \rightarrow z_0} \frac{d^{N-1}}{dz^{N-1}} [f(z)(z-z_0)^N]}$$

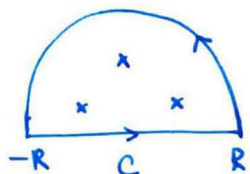
Residue theorem

If $f(z)$ is analytic in an area bounded by a contour C , except for a finite number of poles & isolated essential singularities, then

$$\oint f(z) dz = 2\pi i \sum \text{Residues}$$

Example: $\int_0^{\infty} \frac{x^2 dx}{x^6+1} = ?$

$$I = \int_0^{\infty} \frac{x^2 dx}{x^6+1} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2 dx}{x^6+1}$$


$$\oint \frac{z^2 dz}{z^6+1} = 2I + \int_{\Gamma} \frac{z^2 dz}{z^6+1}$$
$$I_2 = \int_C \frac{z^2 dz}{z^6+1}, \quad I_0 = \oint \frac{z^2 dz}{z^6+1}$$

$$|I_2| \leq \int_{\Gamma} \frac{|z|^2 |dz|}{|z^6+1|} = R^2 \int_{\Gamma} \frac{|dz|}{|z^6+1|} \leq \frac{R^2 \pi R}{\min |z^6+1|}$$

~~$$|z^6+1| > ||z|^6-1| = R^6-1$$~~

$$|z^6+1| > ||z|^6-1| = R^6-1$$

$$\Rightarrow |I_2| \leq \frac{\pi R^3}{R^6-1} \rightarrow 0 \text{ as } R \rightarrow \infty$$

$$\Rightarrow I = \frac{1}{2} I_0$$

$$z^6+1 \text{ has simple poles at } z_k = e^{i\pi/6 + 2k\pi i/6} = e^{i\pi/6(2k+1)} \quad k=0,1,2,\dots,5$$

$$\text{Poles inside } C: e^{i\pi/6}, e^{i\pi/2}, e^{5i\pi/6}$$

$$\text{Residue at } z_k = \lim_{z \rightarrow z_k} (z-z_k) \frac{z^2}{z^6+1} = \lim_{z \rightarrow z_k} \frac{3z^2 - 2zz_k}{6z^5} = \frac{1}{6z_k^3}$$

$$\Rightarrow I = \frac{1}{2} \times \frac{2\pi i}{6} \left[\frac{1}{e^{i\pi/2}} + \frac{1}{e^{3i\pi/2}} + \frac{1}{e^{5i\pi/2}} \right] = \frac{\pi}{6}$$

(2) Find residues at all poles in finite complex plane for

(a) $\frac{z^2 - 2z}{(z+1)^2(z^2+4)}$

(b) $e^z \operatorname{cosec}^2 z$

(a) Poles at $\begin{cases} z = -1 & (2\text{nd order}) \\ z = \pm 2i & (\text{Simple}) \end{cases}$

For the simple poles,

$$\text{Residue} = \lim_{z \rightarrow z_0} (z - z_0) \frac{z^2 - 2z}{(z+1)^2(z^2+4)} = \frac{(\pm 2i)^2 - 2(\pm 2i)}{(\pm 2i+1)^2(\cancel{z-z_0} \pm 2i \pm 2i)}$$

$$= \frac{-4 \mp 4i}{(-4+1 \pm 4i)(\pm 4i)} = \frac{(-1 \pm i)}{(-3 \pm 4i)}$$

$$= \frac{(-1 \pm i)(-3 \mp 4i)}{25}$$

$$= \frac{+3+4 \mp 3i \pm 4i}{25} = \frac{7 \pm i}{25}$$

For $z = -1$

$$\text{Residue} = \lim_{z \rightarrow -1} \frac{d}{dz} \left[(z+1)^2 \left(\frac{z^2 - 2z}{(z^2+4)(z+1)^2} \right) \right]$$

$$= \lim_{z \rightarrow -1} \frac{d}{dz} \frac{z^2 - 2z}{z^2 + 4}$$

$$= \lim_{z \rightarrow -1} \frac{(z^2+4)(2z-2) - (z^2-2z)(2z)}{(z^2+4)^2}$$

$$= \frac{-20 - (3)(-2)}{25} = -\frac{14}{25}$$

(b) $f(z) = \frac{e^z}{\sin^2 z}$

~~Simple poles at zeros of $\sin z$~~

Poles of order 2 at zeros of $\sin z$: $z = m\pi$ $m \in \mathbb{Z}$

let $z - m\pi = u$

$$\Rightarrow \sin z = \sin(u + m\pi) = (-1)^m \sin u$$

$$\Rightarrow \sin^2 z = \sin^2 u$$

$$\Rightarrow f(z) = \frac{e^z}{\sin^2 z} = e^{m\pi} e^u (\sin u)^{-2}$$

$$= e^{m\pi} \left(1 + u + \frac{u^2}{2!} + \dots\right) \left(u - \frac{u^3}{3!} + \frac{u^5}{5!} + \dots\right)^{-2}$$

$$= \frac{e^{m\pi}}{u^2} \left(1 + u + \frac{u^2}{2!} + \dots\right) \left(1 + \frac{u^2}{3} + \dots\right)$$

$$= e^{m\pi} \left(\frac{1}{u^2} + \frac{1}{u} + \dots\right)$$

$\Rightarrow$ Residue at $z=m\pi$ is $e^{m\pi}$

③ $\oint_C \frac{dz}{z \sin z}$ C : Circle encircling origin

$$\frac{1}{z \sin z} = \frac{1}{z^2 \left(1 - \frac{z^2}{2!} + \frac{z^4}{5!} - \dots\right)}$$

$$= \frac{1}{z^2} \left[1 + \left(z^2 - \frac{z^4}{5!} + \dots\right) + \left(z^2 - \frac{z^4}{5!} + \dots\right)^2 + \dots \right]$$

$$= \frac{1}{z^2} \left[1 + z^2 + \frac{119}{120} z^4 + \dots \right] = \sum_{n=-\infty}^{\infty} a_n z^n$$

$$\Rightarrow \text{Residue} = a_{-1} = 0$$

$$\Rightarrow \oint \frac{dz}{z \sin z} = 0$$

[Second Method: $\frac{1}{z \sin z}$ is even $\Rightarrow$ For closed circular

contours around origin, for every z , there is a contribution from $-z$ whose contribution cancels the contribution from z to the integral]

④ $\oint_C \frac{e^z dz}{z(z-1)}$ C : Contour of Radius 2 around origin

Simple poles at $z=0$ and $z=1$

Residue at $z=0$: $\lim_{z \rightarrow 0} \frac{e^z}{z-1} = -1$

Residue at $z=1$: $\lim_{z \rightarrow 1} \frac{e^z}{z} = e$

$$\Rightarrow \oint_c \frac{e^z dz}{z(z-1)} = 2\pi i(e-1)$$

Integrals of the form $I = \int_0^{2\pi} d\theta f(\sin\theta, \cos\theta)$

$$z = e^{i\theta} \Rightarrow dz = i e^{i\theta} d\theta \Rightarrow d\theta = \frac{dz}{iz}$$

$$\Rightarrow I = \frac{1}{i} \oint_c \frac{dz}{z} f\left(\frac{z-1/z}{2i}, \frac{z+1/z}{2}\right)$$

c : unit circle around origin

Examples:

$$\begin{aligned} \textcircled{1} I &= \int_0^{2\pi} \frac{d\theta}{2+\cos\theta} = \oint_c \frac{dz/iz}{2 + \frac{z+1/z}{2}} \\ &= \frac{2}{i} \oint \frac{dz}{z^2 + 4z + 1} \end{aligned}$$

Simple poles at $\frac{-4 \pm \sqrt{16-4}}{2} = -2 \pm \sqrt{3}$

Only $z = -2 + \sqrt{3}$ is inside the unit circle

$$\Rightarrow I = 2\pi i \times \frac{2}{i} \times \frac{1}{(-2+\sqrt{3}) - (-2-\sqrt{3})} = \frac{2\pi}{\sqrt{3}}$$

$$(1) \quad \frac{1}{2\pi i} \oint_C \frac{e^{+2t}}{z^2(z^2+2z+2)}$$

$$C: |z|=3$$

$$\text{Ans. } \frac{t-1}{2} + \frac{e^{-t} \cos t}{2}$$

$$(2) \quad \int_0^{\infty} \frac{dx}{x^6+1}$$

$$\text{Ans. } 2\pi/3$$

$$(3) \quad \int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+1)^2(x^2+2x+2)}$$

$$\text{Ans. } \frac{7\pi}{50}$$

$$(4) \quad \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^2}$$

$$\text{Ans. } \pi/2$$

$$(5) \quad \int_0^{2\pi} \frac{d\theta}{3-2\cos\theta+\sin\theta}$$

$$\text{Ans. } \pi$$

$$(6) \quad \int_0^{2\pi} \frac{d\theta}{(5-3\sin\theta)^2}$$

$$\text{Ans. } 5\pi/32$$

Cauchy Principal Value of the real integral of $f(x)$ with an isolated singularity at x_0 is

$$P \int_a^b f(x) dx \text{ or } \oint f(x) dx \equiv \lim_{\delta \rightarrow 0^+} \left[\int_a^{x_0-\delta} f(x) dx + \int_{x_0+\delta}^b f(x) dx \right]$$

Example: $I = \int_0^{\infty} \frac{\sin x}{x} dx$

$$= \int_0^{\infty} \frac{e^{ix} - e^{-ix}}{2ix} dx$$

$$= \frac{1}{2i} \int_0^{\infty} \left[\frac{e^{ix}}{x} + \frac{e^{-ix}}{-x} \right] dx$$

$$= \frac{1}{2i} P \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx$$

$$\oint \frac{e^{iz}}{z} dz = 0$$

$$= 2iI + I_s + I_R$$

$$I_s = \lim_{\delta \rightarrow 0} \int_{-\pi}^{\pi} \frac{e^{i\delta e^{i\theta}}}{\delta e^{i\theta}} d(\delta e^{i\theta}) = \lim_{\delta \rightarrow 0} \int_{-\pi}^{\pi} e^{i\delta e^{i\theta}} i d\theta$$

$$= -i \lim_{\delta \rightarrow 0} \int_0^{\pi} e^{i\delta e^{i\theta}} d\theta$$

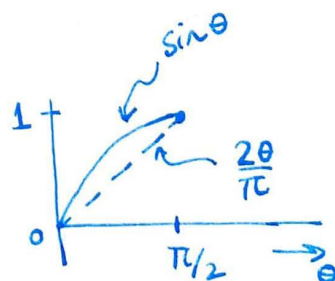
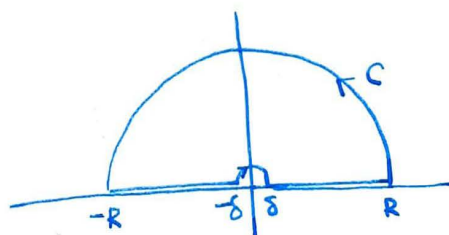
$$= -i \int_0^{\pi} d\theta = -i\pi$$

$$I_R = \lim_{R \rightarrow \infty} \int_0^{\pi} \frac{e^{iRe^{i\theta}}}{Re^{i\theta}} R i e^{i\theta} d\theta$$

$$= i \lim_{R \rightarrow \infty} \int_0^{\pi} e^{iR \cos \theta - R \sin \theta} d\theta$$

$$|I_R| \leq \lim_{R \rightarrow \infty} \int_0^{\pi} e^{-R \sin \theta} d\theta \leq \lim_{R \rightarrow \infty} \int_0^{\pi} e^{-R 2\theta/\pi} d\theta$$

$$\Rightarrow |I_R| \leq \lim_{R \rightarrow \infty} \left[\frac{-\pi}{2R} e^{-2R\theta/\pi} \right]_0^{\pi} = 0 \Rightarrow I_R = 0$$



$$2iI - i\pi = 0 \Rightarrow I = \pi/2 \Rightarrow$$

$$\boxed{\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}}$$

Jordan's lemma:

If $f(Re^{i\theta}) \rightarrow 0$ as $R \rightarrow \infty$ and $0 \leq \theta \leq \pi$, then

$$I_R \equiv \lim_{R \rightarrow \infty} \int_C e^{iaz} f(z) dz = 0$$

where a is a positive real number and C is a semicircular contour of radius R centered at origin

$$I_R = \lim_{R \rightarrow \infty} \int_0^{\pi} e^{iaR\cos\theta - aR\sin\theta} f(Re^{i\theta}) Rie^{i\theta} d\theta$$

$$\Rightarrow |I_R| \leq \lim_{R \rightarrow \infty} \int_0^{\pi} e^{-aR\sin\theta} |f(Re^{i\theta})| R d\theta$$

$$\leq \lim_{R \rightarrow \infty} R E_R \int_0^{\pi} e^{-aR\sin\theta} d\theta$$

$$\leq \lim_{R \rightarrow \infty} R E_R \int_0^{\pi} e^{-2aR\theta/\pi} d\theta$$

$$= \lim_{R \rightarrow \infty} R E_R \left[\frac{e^{-2aR\theta/\pi}}{-2aR/\pi} \right]_0^{\pi}$$

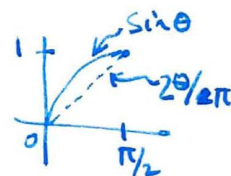
$$= \lim_{R \rightarrow \infty} \frac{\pi E_R}{2a} [1 - e^{-2aR}]$$

$$= 0$$

$$\because \lim_{R \rightarrow \infty} E_R = 0$$

$$\Rightarrow I_R = 0$$

where $E_R = \max_{\theta} |f(Re^{i\theta})|$

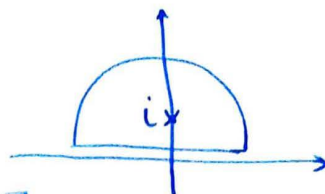


$$\textcircled{1} \int_{-\infty}^{\infty} dx \frac{\cos x}{x^2+1} = \operatorname{Re} \int_{-\infty}^{\infty} dx \frac{e^{ix}}{x^2+1}$$

$$= \operatorname{Re} \left[\oint_C dz \frac{e^{iz}}{z^2+1} - I_R \right]$$

$$= \operatorname{Re} \left[2\pi i \frac{e^{i(i)}}{i+i} \right]$$

$$= \frac{\pi}{e}$$



$I_R \rightarrow 0$ by Jordan's lemma

$$\textcircled{2} \int_0^{\infty} \frac{x dx}{x^4+1} = I_1 + I_2 + I_R$$

$$I_{\text{total}} = \oint \frac{z dz}{z^4+1} = I_1 + I_2 + I_R$$

$$I_{\text{total}} = 2\pi i (\text{Residue at } e^{i\pi/4})$$

$$= 2\pi i \frac{e^{i\pi/4}}{2i \times 2e^{i\pi/4}} = \frac{\pi}{2}$$

$$I_R = \lim_{R \rightarrow \infty} \int R i e^{i\theta} d\theta \frac{R e^{i\theta}}{R e^{4i\theta} + 1}$$

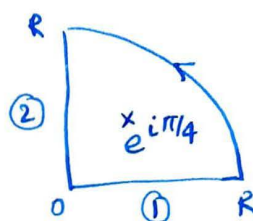
$$|I_R| \leq \lim_{R \rightarrow \infty} R^2 \int \frac{d\theta}{|R e^{4i\theta} + 1|}$$

$$\leq \lim_{R \rightarrow \infty} R^2 \int \frac{d\theta}{R-1} \rightarrow 0 \quad \text{as } I_R \rightarrow 0$$

$$\Rightarrow I_R = 0$$

$$I_2 = \int_0^{\infty} \frac{iy idy}{y^4+1} = \int_0^{\infty} \frac{y dy}{y^4+1} = I_1$$

$$\Rightarrow 2I_1 = \pi/2 \Rightarrow I_1 = \frac{\pi}{4}$$



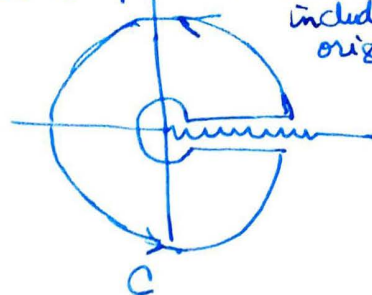
$$\begin{aligned} z^4+1 &= (z^2+i)(z^2-i) \\ &= (z^2+i)(z+e^{i\pi/4})(z-e^{i\pi/4}) \end{aligned}$$

Integrals of the form $\int_0^{\infty} dx \frac{P(x)}{Q(x)}$

Degree of Q at least 2 more than P & Q has no zeros on positive real axis including origin

Consider $J(E, R) = \oint_C dz \frac{P(z) \log z}{Q(z)}$

Contributions from circular parts $\rightarrow 0$



$$\lim_{\eta \rightarrow 0} [\log(x+i\eta) - \log(x-i\eta)] = -2\pi i$$

$$\Rightarrow \boxed{J(0, \infty) = -2\pi i I}$$

Example $\int_0^{\infty} \frac{dx}{(x+1)(x+2)(x+3)}$

Ans. $\ln 2 - \frac{1}{2} \ln 3$

① $\int_0^{\infty} \frac{\ln x dx}{(x^2+4)^2} = \frac{\pi}{32} (\ln 2 - 1)$



② $\int_0^{\infty} \frac{x^{-a}}{1+x} dx = \frac{\pi}{\sin a\pi}$

$0 < a < 1$



Laplace Transform

Some functions do not have convergent Fourier transforms.

e.g., $f(x) = 1$, $f(x) = x$, etc.

Sometimes, multiplication by e^{-cx} [$c \in \mathbb{R}$ and is greater than some constant depending on the function that was chosen] gets rid of the problem at $+\infty$. It however introduces additional problems at $-\infty$.

Consider the function $f(x)e^{-cx} \Theta(x)$ where $\Theta(x)$ is the Heaviside step function. $\Theta(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$

This function may have a convergent Fourier transform.

$$\begin{aligned} F_1(k) &= \int_{-\infty}^{\infty} f(x) e^{-cx} \Theta(x) e^{-ikx} dx \\ &= \int_0^{\infty} f(x) e^{-(c+ik)x} dx \quad \dots (1) \end{aligned}$$

Its inverse Fourier transform is,

$$f(x) e^{-cx} \Theta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F_1(k) e^{ikx} dk \quad \dots (2)$$

We make a change of variables: $s = c + ik$, and write

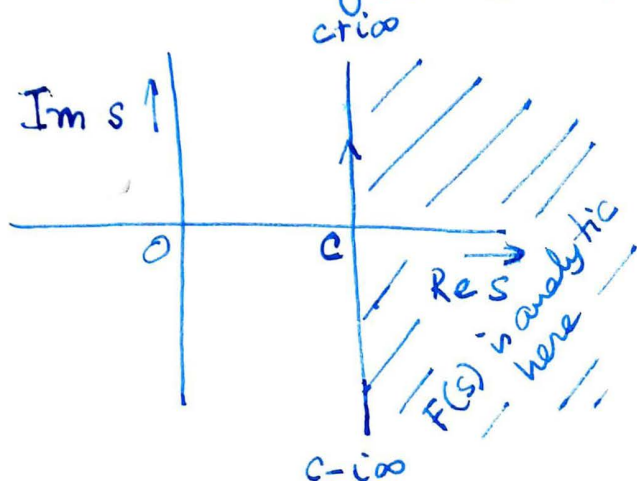
$F(s) = F_1\left(\frac{s-c}{i}\right) = F_1(k)$ to get the Laplace transform of $f(x)$.

$$F(s) = \int_0^{\infty} f(x) e^{-sx} dx \quad \dots (3)$$

To get the inverse Laplace transform, we multiply Eq.(2) by e^{cx} and make a change of variables, $s = c + ik \Rightarrow k = \frac{s-c}{i}$

Thus, $f(x) \mathcal{H}(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s) e^{sx} ds.$

This integral is known as the Bromwich integral.



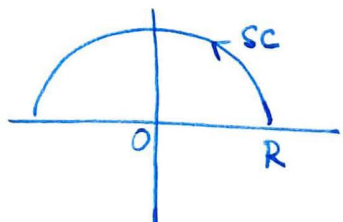
c is chosen so that $F(s)$ is analytic for $\text{Re } s > c$.

Aside

Other forms of Jordan's lemma:

① Usual Form

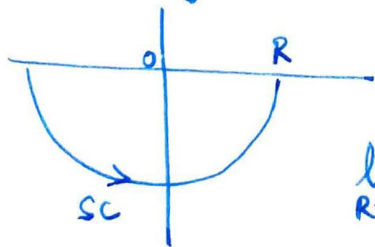
If $\lim_{R \rightarrow \infty} f(Re^{i\theta}) = 0$ for $0 \leq \theta \leq \pi$,



$$\lim_{R \rightarrow \infty} \int_{sc} f(z) e^{iaz} dz = 0 \quad a \in \mathbb{R} \text{ and } a > 0$$

② [Change variables $z = -t$]

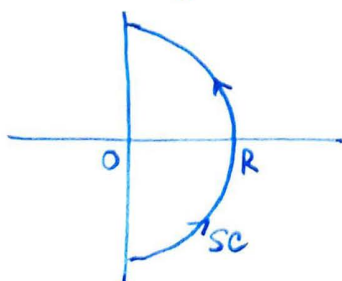
If $\lim_{R \rightarrow \infty} f(Re^{i\theta}) = 0$ for $\pi \leq \theta \leq 2\pi$



$$\lim_{R \rightarrow \infty} \int_{sc} f(z) e^{-iaz} dz = 0, \quad a \in \mathbb{R}, a > 0.$$

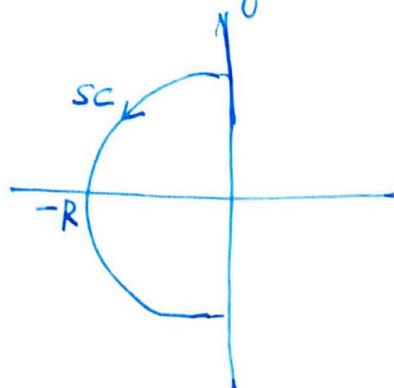
③ [Change variables $z = +it$]

If $\lim_{R \rightarrow \infty} f(Re^{i\theta}) = 0$ for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$



$$\lim_{R \rightarrow \infty} \int_{sc} f(z) e^{-az} dz = 0, \quad a \in \mathbb{R}, a > 0.$$

④ [Change variables $z = -it$]



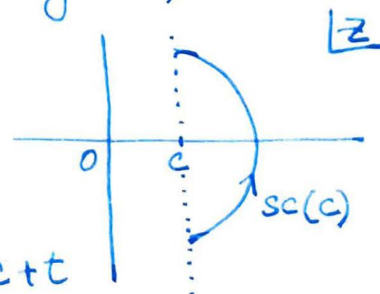
If $\lim_{R \rightarrow \infty} f(Re^{i\theta}) = 0$ for $\pi/2 \leq \theta \leq 3\pi/2$

$$\lim_{R \rightarrow \infty} \int_{sc} f(z) e^{az} dz = 0, \quad a \in \mathbb{R}, a > 0$$

⑤ In all of the above results, you could shift the origin to an arbitrary point z_0 in the complex plane ($|z_0| < \infty$)

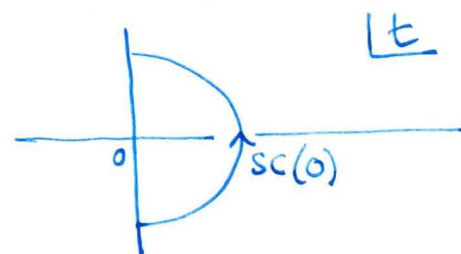
e.g. ~~from ③~~, consider the integral,

$$\lim_{R \rightarrow \infty} \int_{sc(c)} f(z) e^{-az} dz$$



$$= \lim_{R \rightarrow \infty} \int_{sc(0)} f(c+t) e^{-a(c+t)} dt \quad z = c+t$$

$$= e^{-ac} \lim_{R \rightarrow \infty} \int_{sc(0)} f(c+t) e^{-at} dt$$



$$= 0 \quad \text{if} \quad f(c+Re^{i\theta}) \rightarrow 0 \quad \text{as} \quad R \rightarrow \infty \quad \text{for} \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

Note: The above results are also valid for any finite fraction of the semi-circle, e.g., a quarter circle.

Example 1.

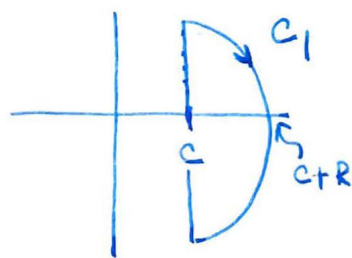
Find the function whose Laplace transform is $F(s) = \frac{1}{s}$.

$F(s)$ has a simple pole at the origin. We can therefore choose the constant c to be any positive real number.

We use the Bromwich integral,

$$f(x) \oplus (x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s) e^{xs} ds \quad \dots (1)$$

For $x < 0$, we use the contour C_1 .



$$\oint_{C_1} F(s) e^{xs} ds = 0 \quad \left[\because \text{there are no singularities of } F(s) \text{ inside } C_1 \right]$$

$$\Rightarrow \int_{c-i\infty}^{c+i\infty} F(s) e^{xs} ds + I_{sc_1} = 0$$

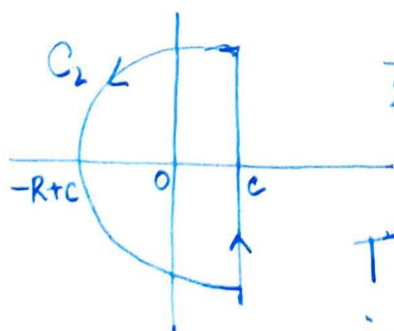
$I_{sc_1} = 0$ from Jordan's lemma since,

$$\lim_{R \rightarrow \infty} F(c + Re^{i\theta}) = \lim_{R \rightarrow \infty} \frac{1}{c + Re^{i\theta}} = 0$$

for $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$\Rightarrow$ Both sides of Eq (1) are zero for $x < 0$ (as expected).

For $x > 0$, we close the contour on the opposite side



$$\begin{aligned} \frac{1}{2\pi i} \oint_{C_2} F(s) e^{xs} ds &= \frac{1}{2\pi i} \times 2\pi i \times (\text{Residue at } s=0) \\ &= \lim_{s \rightarrow 0} s \times \frac{e^{xs}}{s} = 1 \end{aligned}$$

The contribution from the semi-circular part is zero because of Jordan's lemma (Check.)

$$\Rightarrow f(x) = 1.$$

[Check this result by Laplace transforming $f(x) = 1$.]

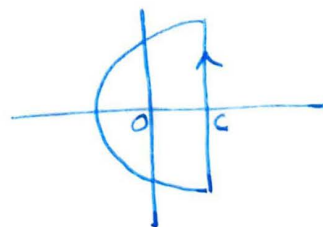
Example 2.

$$F(s) = \frac{1}{s-z} \quad \text{where } z \text{ is a complex number, } |z| < \infty$$

Choose $c > \operatorname{Re} z$

Check the expected result for $x < 0$.

For $x > 0$ we use the same contour as in example 1, with $c > \operatorname{Re} z$.



From Jordan's lemma, the semi-circular part does not contribute,

$$\therefore \lim_{R \rightarrow \infty} \frac{1}{c + Re^{i\theta} - z} = 0 \quad \text{for } \pi/2 \leq \theta \leq 3\pi/2$$

$$\begin{aligned} \therefore f(x) &= \frac{1}{2\pi i} \times 2\pi i \times (\text{Residue at } s=z) \\ &= \lim_{s \rightarrow z} (s-z) \frac{e^{xs}}{s-z} = e^{xz} \end{aligned}$$

[FILL IN THE MISSING STEPS AND MAKE SURE YOU UNDERSTAND EVERYTHING.]

(a) For $z = \lambda$, ($\lambda \in \mathbb{R}$), inverse Laplace transform of $\frac{1}{s-\lambda}$ is $e^{\lambda x}$

(b) $z = i\lambda$. Inverse Laplace transform of $\frac{1}{s-i\lambda}$ is $e^{i\lambda x}$.

Since Laplace transform is linear,

$$\mathcal{L}^{-1} \left[\operatorname{Re} \frac{1}{s - i\lambda} \right] = \operatorname{Re} e^{i\lambda x}$$

$$\Rightarrow \mathcal{L}^{-1} \left[\frac{s}{s^2 + \lambda^2} \right] = \cos \lambda x$$

and $\mathcal{L}^{-1} \left[\operatorname{Im} \frac{1}{s - i\lambda} \right] = \operatorname{Im} e^{i\lambda x}$

$$\Rightarrow \mathcal{L}^{-1} \left[\frac{\lambda}{s^2 + \lambda^2} \right] = \sin \lambda x$$

Find the inverse Laplace transform of $\frac{1}{s^n}$, $n \in \mathbb{Z}, n > 0$.